PHYS 225

Week 15, Lecture 13: Complex numbers and Fourier transforms

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

Reminders

- Last Homework 13 assigned 12/4 due next Wednesday 12/10.
- Last discussion this week 12/5.
- **Final exam:** Tuesday, May 12th at 1:30 – 4:30 PM.
  Location: Noyes Laboratory 100,
  Noyes Laboratory 217.
- ~8 questions (~24 sub-questions) covering material from the entire course (including the first half of the semester!)
- You can bring your formula sheet - 2 double-sided pages. Exam will also have a comprehensive formula sheet.
- If you need DRES/TAC accommodations, set them up by next week
- Bonus assignment due in Gradescope by 11:59 pm on 5/11

Week 15, Lecture 13
Complex numbers and Fourier transforms
Objectives:

- Define the square root of -1, and from that, complex numbers
- Learn the relationship between complex numbers and trig functions (Euler's Identity)
- Add, multiply, and divide complex numbers, and convert between Cartesian and polar representations
- Solve partial differential equations by Fourier transforming

One Minute Paper: in what areas of physics or math have you encountered complex numbers before, or is this your first time being introduced to them?

Week 15, Lecture 13
The imaginary unit

It's annoying when equations don't have solutions.

$$x^2 - 4 = 0 \implies x = \pm 2$$

$$x^2 + 1 = 0 \implies x = ??$$

No ordinary number satisfies this equation, because squares of numbers are positive. So let's boldly define:

$$i \equiv \sqrt{-1}$$ the imaginary unit

$$x^2 + 1 = 0 \implies x = \pm i$$

Week 15, Lecture 13
Complex numbers

That would be overkill if we introduced a totally new concept just to solve one equation. But it does more than that…

Fundamental theorem of algebra: **all** polynomial equations have solutions that are linear combinations of ordinary ("real") numbers and the imaginary unit. Such numbers are called complex numbers

So we don't just get to solve one equation, we get to solve infinitely many equations!

real part

$$\underbrace{z}_{\text{complex number}} = \overbrace{x}^{\text{real part}} + i\underbrace{y}_{\text{imaginary part}}$$

complex number

imaginary part

Week 15, Lecture 13
Manipulating complex numbers

Just like real numbers, we can add, subtract, multiply, and divide complex numbers. (Though we should still avoid dividing by 0.)

$$z_1 = 3 + 2i$$
$$z_2 = 1 + 5i$$

$$z_1 + z_2 = (3 + 2i) + (1 + 5i)$$
$$= (3 + 1) + (2 + 5)i$$
$$= 4 + 7i$$

$$z_1 - z_2 = (3 + 2i) - (1 + 5i)$$
$$= (3 - 1) + (2 - 5)i$$
$$= 2 - 3i$$

$$z_1 z_2 = (3 + 2i)(1 + 5i)$$
$$= 3(1) + 3(5i) + 2i(1) + 2i(5i)$$
$$= (3 + 10i^2) + (15 + 2)i = -7 + 17i$$

Week 15, Lecture 13
Manipulating complex numbers

Having complex numbers in the denominator of a fraction is fine, but annoying if we want to figure out the real and imaginary parts. The trick is to "realify" the denominator, analogous to rationalizing the denominator when there are square roots.

$$\frac{z_1}{z_2} = \frac{(3+2i)}{(1+5i)} = \frac{(3+2i)}{(1+5i)}\frac{(1-5i)}{(1-5i)}$$

$$= \frac{(3+2i)(1-5i)}{(1-25i^2)} = \frac{(3+10)+(2-15)i}{26} = \frac{1}{2}+\frac{1}{2}i$$

Week 15, Lecture 13
Complex conjugation

That denominator trick is so useful that it gets its own name: **complex conjugation.**

$$z = x + iy$$

$$z^* \equiv \bar{z} \equiv x - iy$$

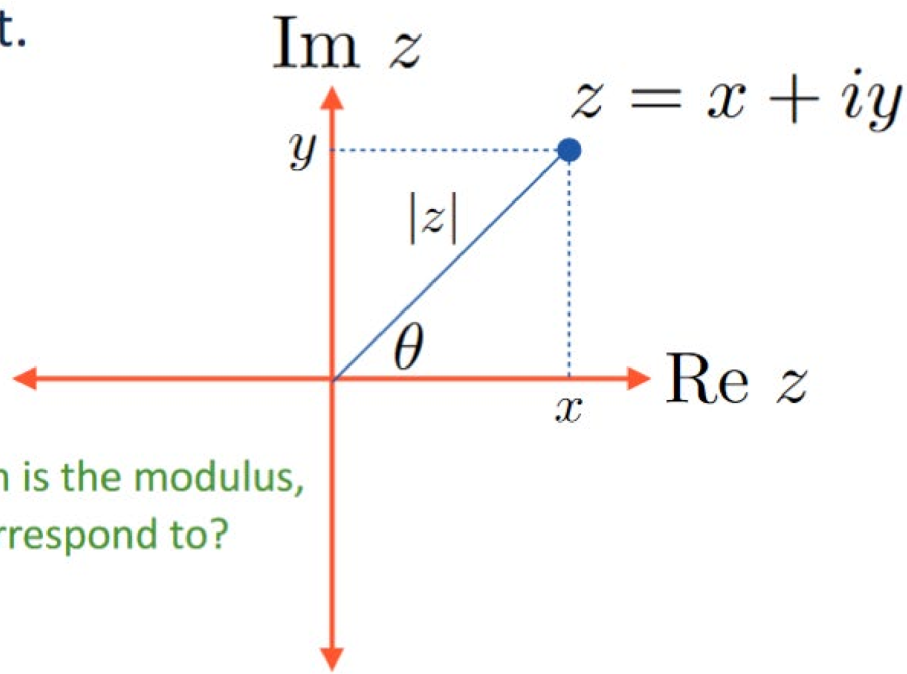
different conventions, you'll see both in books; they mean the same thing

To take the complex conjugate of an expression involving complex numbers, replace **all** instances of i with -i.

Multiplying a complex number by its conjugate and taking the square root gives us a non-negative real number, called the **modulus** or **magnitude** of z: $|z| \equiv \sqrt{z^* z} = \sqrt{x^2 + y^2}$

Week 15, Lecture 13
The complex plane

Since complex numbers are represented by two real numbers (the real and imaginary parts), we can try graphing them in the **complex plane**, where the x-axis is the real part and the y-axis is the imaginary part.



The distance from the origin is the modulus, but what does the angle correspond to?

Week 15, Lecture 13
Euler's Identity

Let's take |z| = 1. Then from the complex plane diagram we can see that $x = \cos\theta$ and $y = \sin\theta$

$$z = x + iy = \cos\theta + i\sin\theta$$

A trick (not obvious!): let's write the right-hand side as a Taylor series

[blackboard]

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots$$

$$\cos\theta + i\sin\theta = 1 + i\theta - \frac{\theta^2}{2!} - \frac{i\theta^3}{3!} + \frac{\theta^4}{4!} + \cdots$$

Note that $(i\theta)^2 = i^2\theta^2 = -\theta^2$, and

$$(i\theta)^3 = i^3\theta^3 = i(i^2)\theta^3 = -i\theta^3$$

$$(i\theta)^4 = i^4\theta^4 = (i^2)^2\theta^4 = (-1)^2\theta^4 = \theta^4$$

So the RHS is $1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \cdots$

This is the Taylor series for e^x, with $x = i\theta$

Week 15, Lecture 13
Euler's Identity

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[blackboard]

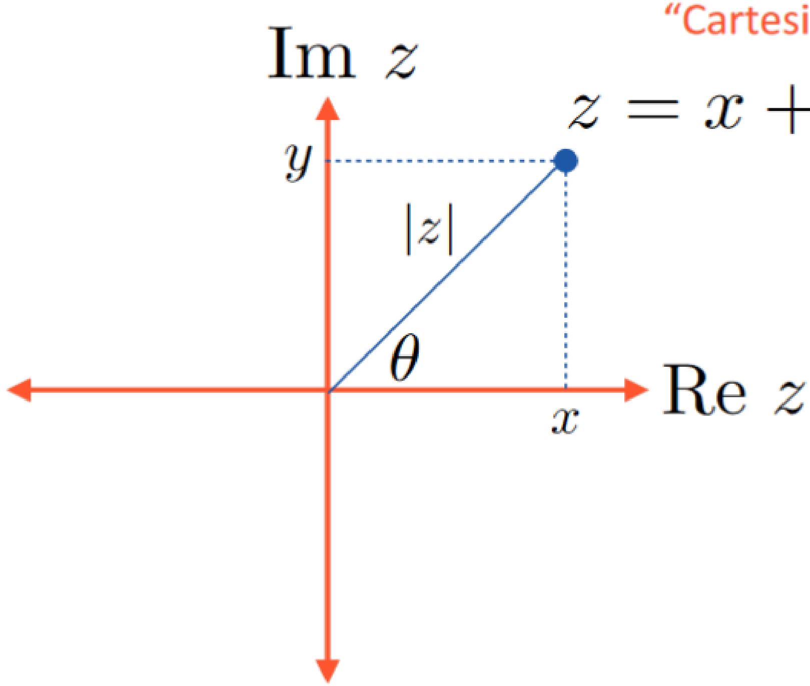
$$e^{i\theta} = \cos\theta + i\sin\theta$$

Taking $\theta = \pi$, we get one of the most famous equations in all of math:

$$e^{i\pi} = -1$$

Week 15, Lecture 13

Polar and Cartesian forms



"Cartesian form"

$$z = x + iy$$

Now that we know that

$$e^{i\theta} = \cos\theta + i\sin\theta$$

we can see that

$$x = |z|\cos\theta,\ \ y = |z|\sin\theta$$

imply we can write z as

$$z = |z|e^{i\theta}$$

Often this is written as $z = re^{i\theta}$ "polar form"

r is called the **modulus**, and θ is called the **argument** or **phase**

Week 15, Lecture 13
Polar and Cartesian forms

Polar form is great for:

- complex conjugation

$$z = re^{i\theta} \implies z^* = re^{-i\theta}$$

- taking fractional powers

$$i = e^{i\pi/2}$$

$$i^{1/4} = (e^{i\pi/2})^{1/4} = e^{i\pi/8} = \cos(\pi/8) + i\sin(\pi/8)$$

- deriving trig identities [blackboard]

Consider squaring both sides of

$$e^{i\theta} = \cos\theta + i\sin\theta$$

$$(e^{i\theta})^2 = (\cos\theta + i\sin\theta)^2$$

$$(e^{i\theta})^2 = e^{2i\theta} = \cos(2\theta) + i\sin(2\theta)$$

$$(\cos\theta + i\sin\theta)^2 = \cos^2\theta + i^2\sin^2\theta + 2i\cos\theta\sin\theta = (\cos^2\theta - \sin^2\theta) + i(2\sin\theta\cos\theta)$$

Real and imaginary parts must each match:

$$\cos 2\theta = \cos^2\theta - \sin^2\theta$$

$$\sin 2\theta = 2\sin\theta\cos\theta$$

Week 15, Lecture 13
Intro to Fourier transforms

Let’s use Euler’s formula to solve differential equations!

Week 15, Lecture 13
Intro to Fourier transforms

Suppose we have a partial differential equation, like the wave equation in one dimension:

$$\frac{\partial^2 f(t,x)}{\partial t^2} = v^2 \frac{\partial^2 f(t,x)}{\partial x^2}$$

From PHYS 212 we know a solution is a periodic function $f = \sin(\omega t + kx)$

We are going to guess a solution: $f = Ae^{i\omega t}e^{ikx}$

Here, A, ω, and k are **constants**: we don't know what they are yet

Week 15, Lecture 13
Intro to Fourier transforms

$$f = Ae^{i\omega t}e^{ikx}$$

Now plug our guess into the wave equation

$$\frac{\partial^2 f(t,x)}{\partial t^2} = v^2 \frac{\partial^2 f(t,x)}{\partial x^2}$$

Our guess turned derivatives into multiplication

$$\Longrightarrow \cancel{A}(i\omega)^2 \cancel{e^{i\omega t}}\cancel{e^{ikx}} = v^2 \cancel{A}(ik)^2 \cancel{e^{i\omega t}}\cancel{e^{ikx}}$$

$$\Longrightarrow \omega^2 = v^2 k^2 \quad (*)$$

So our guess solves the equation as long as ω and k are related by the **dispersion relation** (*)

Week 15, Lecture 13
Intro to Fourier transforms

Why is this a reasonable thing to do?
Linear partial differential equations obey the principle of superposition: a sum of solutions is still a solution.

In fact, so is an infinite sum (integral)!

$$f(x,t) = \int_{-\infty}^{\infty} dk\, \tilde{f}(k) e^{ikx} e^{ivkt}$$

The set of coefficients $\tilde{f}(k)$ can be treated as a function of k, and is called the **Fourier transform** of f

Week 15, Lecture 13
Intro to Fourier transforms

This trick works even better in more dimensions:

$$(\nabla^2 + C)f = 0$$

Laplacian operator $\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \equiv \partial_j \partial^j$

Guess a solution $f = Ae^{ik_x x} e^{ik_y y} e^{ik_z z}$

$$\Longrightarrow \; -k_x^2 - k_y^2 - k_z^2 + C = 0$$

Note minus sign! Each derivative also brings down a factor of i, so two derivatives give $i^2 = -1$

Week 15, Lecture 13
Intro to Fourier transforms

This is a very rich subject which you will doubtless see many, many more times in your physics career. Just a few final remarks:

- You can have either a real or complex solution
  - E&M: Electromagnetic fields are always real
  - Quantum mechanics: Wavefunction is complex

- We guess solutions e^{ikx} when we want the modulus of the guess to be constant.
- We guess solutions e^{kx} when we want exponentially growing or decaying solutions.

Week 15, Lecture 13
Complex numbers and Fourier transforms

Summary

- Defining the imaginary unit as the square root of -1 lets us construct complex numbers with real and imaginary parts
- Complex numbers have a modulus and a phase, which are related to the real and imaginary parts in the same way as polar coordinates are related to Cartesian coordinates
- To divide complex numbers, realify the denominator using the complex conjugate
- When faced with a partial differential equation, try Fourier transforming (i.e. guessing a solution with complex exponentials)

One Minute Paper: write i in polar form.